

AMATH 483 / 583 (Roche) - Homework Set 5

Due Friday May 9, 5pm PT

May 2, 2025

Homework 5 (80 points)

1. (+20) Ring. $\forall x, y \in \mathbb{Z}$ define binary operator $\oplus$ as $x \oplus y = x + y - 1$, and binary operator $\odot$ as $x \odot y = x + y - xy$. $(\mathbb{Z}, \oplus, \odot)$ is a ring.

(a) Additive Identity. Find $z \in \mathbb{Z} \ni a \oplus z = z \oplus a = a, \forall a \in \mathbb{Z}$.

(b) Additive Inverse. $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z} \ni a \oplus b = b \oplus a = z$. Find b .

(c) Show that $\oplus$ commutes.

(d) Multiplicative Identity. A ring with unity is a ring with member $u \in \mathbb{Z}, u \neq z \ni a \odot u = u \odot a = a, \forall a \in \mathbb{Z}$. Find u .

(a)

$$a \oplus z = a + z - 1$$

$$z \oplus a = z + a - 1$$

$$a \oplus z = z \oplus a = a$$

$$a + z - 1 = a$$

$$z - 1 = 0$$

$$\boxed{z = 1}$$

(b)

$$a \oplus b = a + b - 1$$

$$b \oplus a = b + a - 1$$

$$a \oplus b = b \oplus a = z$$

$$z = 1$$

$$a + b - 1 = 1$$

$$a + b = 2$$

$$\boxed{b = 2 - a}$$

$$\text{check: } b + a - 1 = 1$$

$$2 - a + a - 1 = 1$$

$$1 = 1$$

(c)

$$a \oplus b = a + b - 1 = b + a - 1 = b \oplus a$$

$$a \oplus b = b \oplus a$$

This shows $\oplus$ commutes.

(d)

$$a \odot u = a + u - au$$

$$u \odot a = u + a - ua$$

$$a \odot u = u \odot a = a$$

$$a + u - au = a$$

$$(1 - a)u = 0$$

$$\boxed{u = 0}$$

$$\text{check: } u + a - ua = a$$

$$u = 0$$

$$a = a$$

2. (+10) Recursive sequence. Let $T_0 = 2$, $T_1 = 3$, and $T_2 = 6$, and for $n \geq 3$ let $T_n = (n+4)T_{n-1} - 4nT_{n-2} + (4n-8)T_{n-3}$.

- (a) Derive a formula of the form $T_n = A_n + B_n$ where $(A_n), (B_n)$ are well-known sequences.
 (b) Use induction to prove your formula works.

(a)

compute the first values of T_n

$$T_0 = 2$$

$$T_1 = 3$$

$$T_2 = 6$$

$$T_3 = 7 \cdot 6 - 4 \cdot 3 \cdot 3 + (4 \cdot 3 - 8) \cdot 2 = 42 - 36 + 8 = 14$$

$$T_4 = 8 \cdot 14 - 4 \cdot 4 \cdot 6 + (4 \cdot 4 - 8) \cdot 3 = 40$$

$$T_5 = 9 \cdot 40 - 4 \cdot 5 \cdot 14 + (4 \cdot 5 - 8) \cdot 6 = 152$$

$$T_6 = 10 \cdot 152 - 4 \cdot 6 \cdot 40 + (4 \cdot 6 - 8) \cdot 14 = 784$$

$$2^n = 1, 2, 4, 8, 16, 32, \dots$$

$$T_n - 2^n = 1, 1, 2, 6, 24, 120$$

$$n! = 1, 1, 2, 6, 24, 120$$

$$T_n - 2^n = n!$$

$$T_n = 2^n + n!$$

(b)

$$\begin{cases} T_0 = 2, T_1 = 3, T_2 = 6 \\ T_n = (n+4)T_{n-1} - 4nT_{n-2} + (4n-8)T_{n-3}, \quad n \geq 3 \end{cases}$$

$$n=0 \quad T_0 = 2^0 + 0! = 1+1=2$$

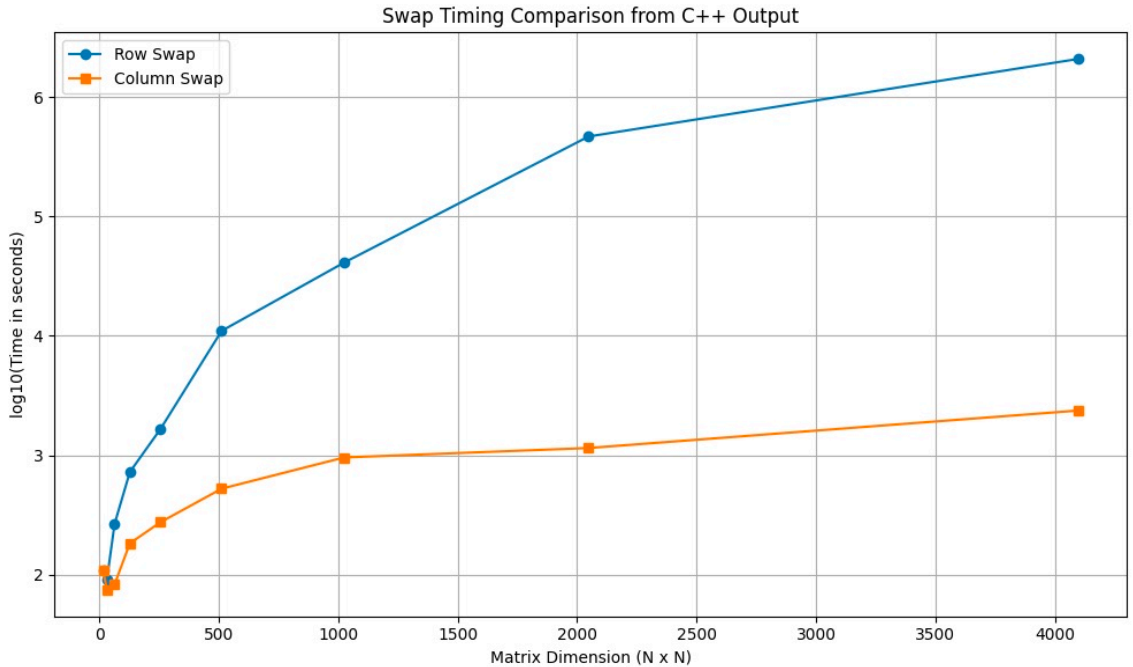
$$n=1 \quad T_1 = 2^1 + 1! = 2+1=3$$

$$n=2 \quad T_2 = 2^2 + 2! = 4+2=6$$

$$\begin{aligned} T_n &= (n+4)T_{n-1} - 4nT_{n-2} + (4n-8)T_{n-3} \\ &= (n+4)(2^{n-1} + (n-1)!) - 4n(2^{n-2} + (n-2)!) + (4n-8)(2^{n-3} + (n-3)!) \\ &= (n+4)2^{n-1} - 4n \cdot 2^{n-2} + (4n-8)2^{n-3} + (n+4)(n-1)! - 4n(n-2)! + (4n-8)(n-3)! \\ &= (n+4) \cdot 2^n - 4n \cdot 2 \cdot 2^{n-3} + (4n-8)2^{n-3} + (n+4)(n-1)(n-2)(n-3)! - 4n(n-2)(n-3)! + (4n-8)(n-3)! \\ &= 8 \cdot 2^{n-3} + (n^3 - 3n^2 + 2n)(n-3)! \\ &= 2^3 \cdot 2^{n-3} + n(n-2)(n-1)(n-3)! \\ &= 2^n + n! \end{aligned}$$

$$T_n = 2^n + n!$$

3. (+20) Memory access time. Write C++ functions that perform **row and column swap** operations on a type double matrix stored in column major index order using a single vector container for the data. The specifications are provided here. Put the functions in file **matrix_swaps.hpp** (no need for header guards -just the code for your functions). Conduct a performance test for square matrix dimensions 16, 32, 64, ... 4096, measuring the time required to conduct row swaps and column swaps separately. Let each operation be measured *ntrial* times, $ntrial \geq 3$. Make a single plot of your *row* and *column* swap timing measurements with time on the y-axis (could be $\log_{10}(time)$) and the problem dimension on the x-axis. Submit files **mem_swaps.hpp** and the plot.



4. (+10) Linear algebra.

- (a) (+5) **Basis.** Does the given set $S = \{1, 1-x, (1-x)^2\}$ form a basis for the space of polynomials up to degree 2, $x \in \mathbb{R}$? Show your work. (Hint: must be both linearly independent and span the space)

1. linearly independent

suppose there are scalars, $\alpha, \beta, \gamma \in \mathbb{R}$

$$\text{Then, } \alpha \cdot 1 + \beta \cdot (1-x) + \gamma (1-x)^2 = 0$$

$$\alpha + \beta - \beta x + \gamma - 2\gamma x + \gamma x^2 = 0$$

$$\alpha + \beta + \gamma - (\beta + 2\gamma)x + \gamma x^2 = 0$$

$$\begin{cases} \gamma = 0 \\ -(\beta + 2\gamma) = 0 \\ \alpha + \beta + \gamma = 0 \end{cases}$$

Hence, the only solution is $\alpha = \beta = \gamma = 0$.

So these three vectors are linearly independent.

2. span the space

use arbitrary quadratic polynomial. $p(x) = ax^2 + bx + c$

$$\alpha \cdot 1 + \beta \cdot (1-x) + \gamma (1-x)^2 = ax^2 + bx + c$$

$$\alpha + \beta + \gamma - (\beta + 2\gamma)x + \gamma x^2 = c + bx + ax^2$$

$$\begin{cases} \gamma = a \\ -(\beta + 2\gamma) = b \\ \alpha + \beta + \gamma = c \end{cases}$$

$$\beta = -b - 2\gamma = -b - 2a$$

$$\alpha = c - \beta - \gamma = c + b + 2a - a = a + b + c$$

$$\begin{cases} \gamma = a \\ \beta = -b - 2a \\ \alpha = a + b + c \end{cases}$$

Thus, for any (a, b, c) , there is a unique solution for (α, β, γ) .

This shows $\{1, 1-x, (1-x)^2\}$ spans.

The set $S = \{1, 1-x, (1-x)^2\}$ forms a basis for the space of polynomials up to degree 2, $x \in \mathbb{R}$ as it is both linearly independent and span the space.

- (b) (+5) **Gram-Schmidt**. Given set $S = \{x_1(t), x_2(t), x_3(t)\} \ni x_1(t) = t^2, \quad x_2(t) = t, \quad x_3(t) = 1$ and $t \in [-1, 1] \in \mathbb{R}$, construct by hand an orthonormal basis for S using the inner product $(x, y) = \int_{-1}^1 x(t)y(t) dt$.

① $v_1 = x_1 = t^2$

$$\begin{aligned}\|v_1\| &= \sqrt{\int_{-1}^1 t^4 dt} \\ &= \sqrt{\frac{t^5}{5} \Big|_{-1}^1} \\ &= \sqrt{\frac{2}{5}}\end{aligned}$$

$$e_1 = \frac{v_1}{\|v_1\|} = \sqrt{\frac{5}{2}} t^2$$

② $x_2 = t$

$$(t, e_1) = \sqrt{\frac{5}{2}} \int_{-1}^1 t^3 dt = 0$$

$$v_2 = x_2 - (t, e_1) \cdot e_1$$

$$= t - 0 \cdot e_1$$

$$v_2 = t$$

$$\|v_2\| = \sqrt{\int_{-1}^1 t^2 dt}$$

$$= \sqrt{\frac{t^3}{3} \Big|_{-1}^1}$$

$$= \sqrt{\frac{2}{3}}$$

$$e_2 = \frac{v_2}{\|v_2\|} = \sqrt{\frac{3}{2}} t$$

③

$$(1, e_1) = \sqrt{\frac{5}{2}} \int_{-1}^1 t^2 dt = \sqrt{\frac{5}{2}} \cdot \frac{2}{3}$$

$$v_3 = 1 - \frac{2}{3} \sqrt{\frac{5}{2}} \cdot e_1 - 0 \cdot e_2$$

$$= 1 - \frac{2}{3} \sqrt{\frac{5}{2}} \cdot \sqrt{\frac{5}{2}} t^2$$

$$= 1 - \frac{5}{3} t^2$$

$$\|v_3\| = \sqrt{\int_{-1}^1 (1 - \frac{5}{3} t^2)^2 dt}$$

$$= \sqrt{\int_{-1}^1 (1 - \frac{10}{3} t^2 + \frac{25}{9} t^4) dt}$$

$$= \sqrt{t - \frac{10}{9} t^3 + \frac{25}{45} t^5 \Big|_{-1}^1}$$

$$= \sqrt{1 - \frac{10}{9} + \frac{25}{45} - (-1 + \frac{10}{9} - \frac{25}{45})}$$

$$= \sqrt{2 - \frac{20}{9} + \frac{50}{45}}$$

$$= \sqrt{\frac{8}{9}}$$

$$= \frac{2\sqrt{2}}{3}$$

$$e_3 = \frac{v_3}{\|v_3\|}$$

$$= (1 - \frac{5}{3} t^2) \cdot \frac{3}{2\sqrt{2}}$$

$$= \frac{3}{2\sqrt{2}} - \frac{5}{2\sqrt{2}} t^2$$

$$= \frac{3 - 5t^2}{2\sqrt{2}}$$

orthonormal basis: $e_1(t) = \sqrt{\frac{5}{2}} t^2$

$$e_2(t) = \sqrt{\frac{3}{2}} t$$

$$e_3(t) = \frac{3 - 5t^2}{2\sqrt{2}}$$

5. (+20) **Strassen**. Use notes from the class lecture to implement the C++ template for the (recursive) Strassen matrix multiplication algorithm for matrices with even dimension, `C = strassenMultiply(A, B);`. Use the provided starter code `strassen.cpp` and implement in it the declared method `strassenMultiply`. Plot the average performance (FLOPs) multiplying `double` precision square matrices of even dimension from $n = 2$ to $n = 512$ as measured $ntrial$ times, $ntrial \geq 3$. Submit `strassen.cpp` and performance plot.

